

Lab 3: Integrating and Differentiating Continuous-Time Signals in MATLAB

Habib University

CE-251L /EE-252L Signal and Systems

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Objective: Evaluating integrals and derivatives of signals using symbolic and numerical methods.

In-Lab:

Part I: Integrating a signal using symbolic and numerical tools.

Part II: Differentiating a signal using symbolic and numerical tools.

Post Lab:

Part III: Integrating and differentiating basic signal functions like unit-impulse and unit-step.

task a

Date . . .

$$y(t) = y(0) + \int_0^t x(\lambda) d\lambda$$

$$x(t) = A \sin(2\pi f_0 t)$$

$$y(t) = y(0) + \int_0^t A \sin(2\pi f_0 \lambda) d\lambda$$

$$= y(0) + \left[\frac{-A \cos(2\pi f_0 \lambda)}{2\pi f_0} \right]_0^t$$

$$= \left[\frac{-1 \cdot \cos(2\pi \lambda)}{2\pi} \right]_0^t$$

$$= \left[\frac{-\cos(2\pi \lambda)}{2\pi} \right]_0^t$$

$$= \left(\frac{-\cos(2\pi t)}{2\pi} \right) - \left(\frac{-\cos(2\pi(0))}{2\pi} \right)$$

$$= -\frac{\cos 2\pi t}{2\pi} + \frac{1}{2\pi}$$

$$= \frac{1 - \cos 2\pi t}{2\pi}$$

$\therefore \sin \frac{\theta}{2} = \frac{1 - \cos \theta}{2}$
 $\therefore \sin \frac{\theta}{2} = \frac{1 - \cos \theta}{2}$

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$= \frac{1 - (1 - 2\sin^2(\pi t))}{2\pi}$$

Date . . .

$$= \frac{2 \sin^2 \pi t}{2\pi}$$

$$= \frac{\sin^2 \pi t}{\pi}$$

to match the answers in matlab

$$\therefore A = 1, f_0 = 1$$

$$= \frac{A \sin^2(f_0 \pi t)}{f_0 \pi}$$

task B

```
syms A f0 t pi
xt = A*sin(2*pi*f0*t)
```

$$xt = A \sin(2 f_0 \pi t)$$

```
yt = int(xt, 't')
```

yt =

$$- \frac{A \cos(2 f_0 \pi t)}{2 f_0 \pi}$$

```
yt = int(xt, 't', [0, t])
```

yt =

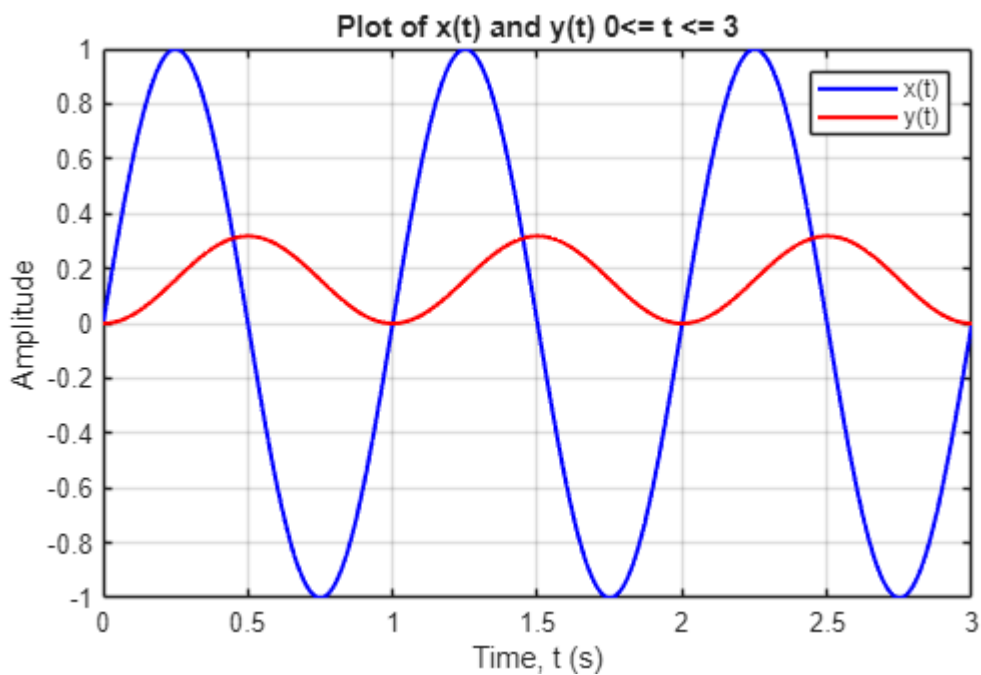
$$\frac{A \sin(f_0 \pi t)^2}{f_0 \pi}$$

observations:

the values that i got in the task b is same as value in task but only because i knew what value i need to get while solving task a otherwise i would not have gotten to that simplified equation

task c

```
clear all;
close all;
clc
A = 1;
f0 = 1;
t = 0:0.01:3;
x = A*sin(2*pi*f0*t);
y = (A*(sin(pi*f0*t)).^2)/(pi*f0);
figure;
plot(t, x, 'b', 'LineWidth', 1.5);
hold on;
plot(t, y, 'r', 'LineWidth', 1.5);
grid on
legend('x(t)', 'y(t)')
xlabel('Time, t (s)');
ylabel('Amplitude');
title('Plot of x(t) and y(t) 0 ≤ t ≤ 3');
```



task d

```
a = [1 2 3 4 5];
b = cumsum(a)
```

```
b = 1×5
    1     3     6    10    15
```

observations:

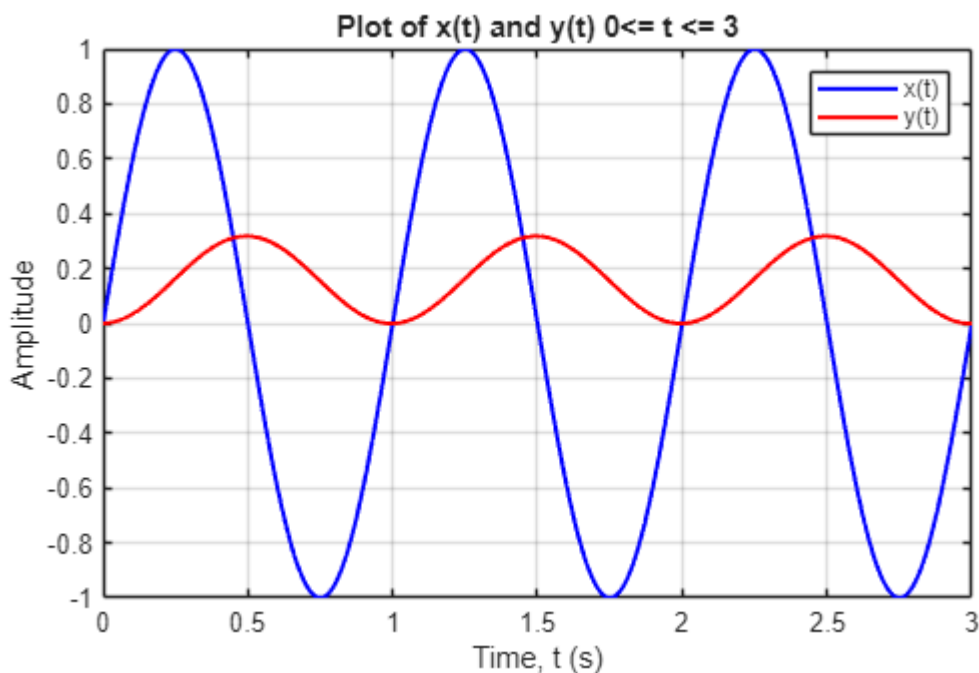
what i observed is that the each element of Array b is the sum of all the element till that index of array a
so it for each element of b it take the cumilative sum of all array a elements till that element

task e

```
clear all;  
close all;  
clc  
A = 1;  
f0 = 1;  
dt = 0.01
```

```
dt =  
0.0100
```

```
t = 0:dt:3;  
x = A*sin(2*pi*f0*t);  
y = dt * cumsum(x);  
figure;  
plot(t, x, 'b', 'LineWidth', 1.5);  
hold on;  
plot(t, y, 'r', 'LineWidth', 1.5);  
grid on  
legend ('x(t)', 'y(t)')  
xlabel('Time, t (s)');  
ylabel('Amplitude');  
title('Plot of x(t) and y(t) 0<= t <= 3');
```



task f

observation

the graph created in task b and task e is same, as we know

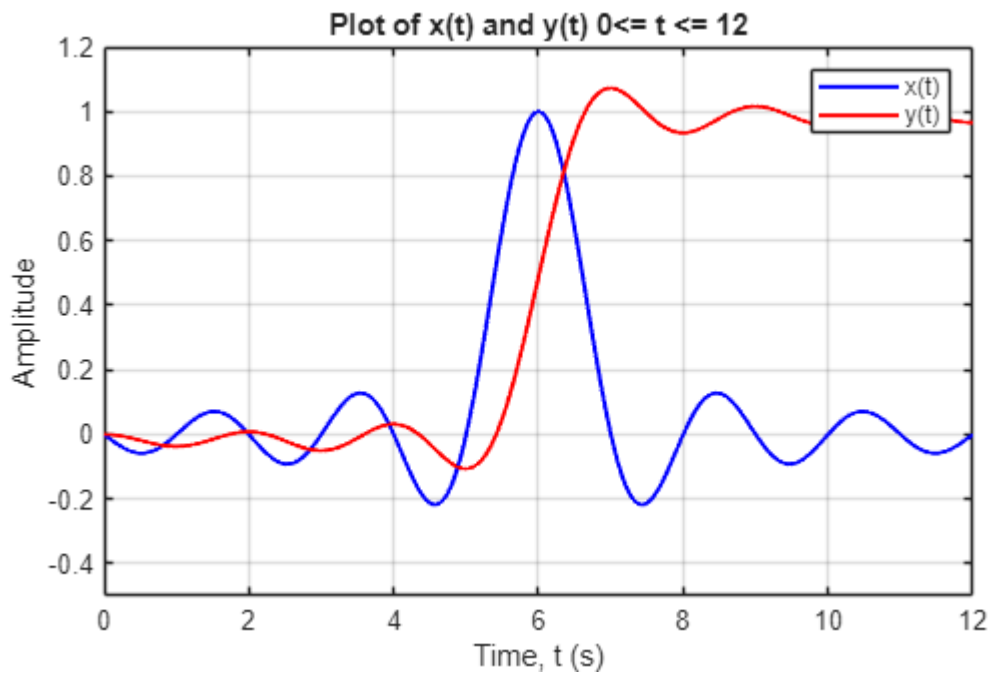
$$y(t) = y(t_0) + \int_{t_0}^t x(\lambda) d\lambda \approx y(t_0) + dt \sum_{k=t_0}^t x[k]$$

task g

by using we numerical integration we can easily solve any equation no matter how complex it would be, while solving by hand there need to be a simplified equation that we hardly would get while working real life problems

task h

```
dt =0.01;
t = 0:dt:12;
x =sinc(t-6);
y = dt * cumsum(x);
figure;
plot(t, x, 'b', 'LineWidth', 1.5);
hold on;
plot(t, y, 'r', 'LineWidth', 1.5);
grid on
legend ('x(t)', 'y(t)')
xlabel('Time, t (s)');
ylabel('Amplitude');
title('Plot of x(t) and y(t) 0<= t <= 12');
axis([0 12 -0.5 1.2])
```



task i

```
a = [1 2 3 4 5]
```

```
a = 1x5
    1    2    3    4    5
```

```
b = diff(a)
```

```
b = 1x4
    1    1    1    1
```

observation:

it gave me a array b and each element of b ($b[i]$) is the the difference of of its corresponding elements in a with the very next elemnent of it ($a[i+1] - a[i]$)

task j

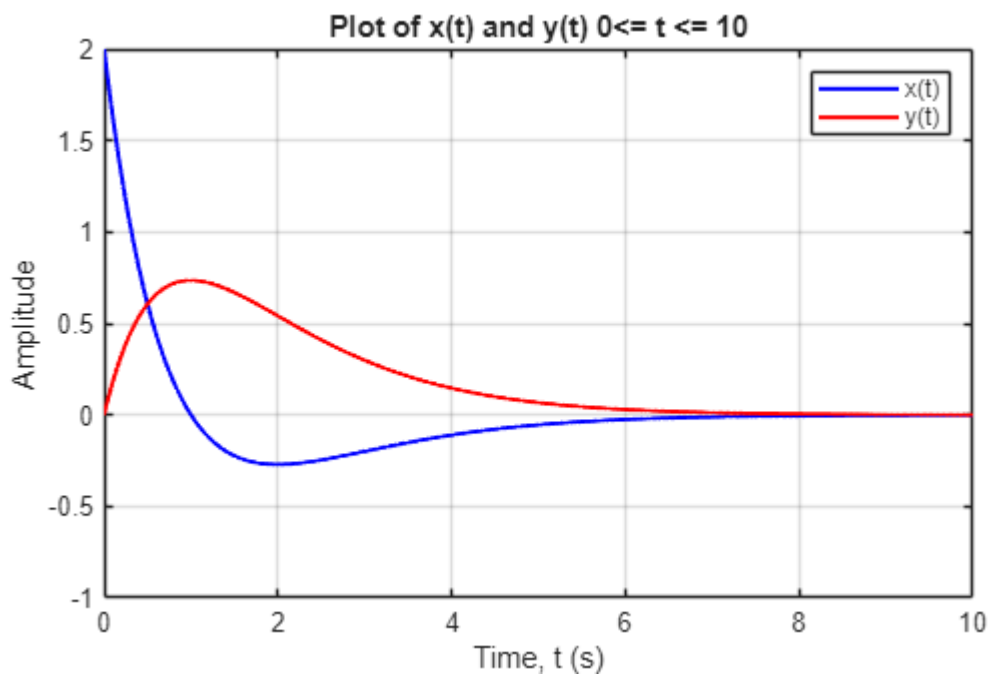
$$\begin{aligned}
 y(t) &= 2t e^{-t} \\
 \frac{d}{dt} (2t e^{-t}) &= \frac{d}{dt} (2t e^{-t}) \\
 \therefore (u \cdot v)' &= u'v + v'u \\
 &= (2)(e^{-t}) + (-e^{-t})(2t) \\
 &= 2e^{-t} - 2te^{-t} \\
 &= 2e^{-t}(1-t)
 \end{aligned}$$

task k

```

dt = 0.01;
t = 0:dt:10;
y = 2.*t.*exp(-t);
x = 2.*exp(-t).*(1-t);
figure;
plot(t, x, 'b', 'LineWidth', 1.5);
hold on;
plot(t, y, 'r', 'LineWidth', 1.5);
grid on
legend('x(t)', 'y(t)')
xlabel('Time, t (s)');
ylabel('Amplitude');
title('Plot of x(t) and y(t) 0 <= t <= 10');
axis([0 10 -1 2])

```

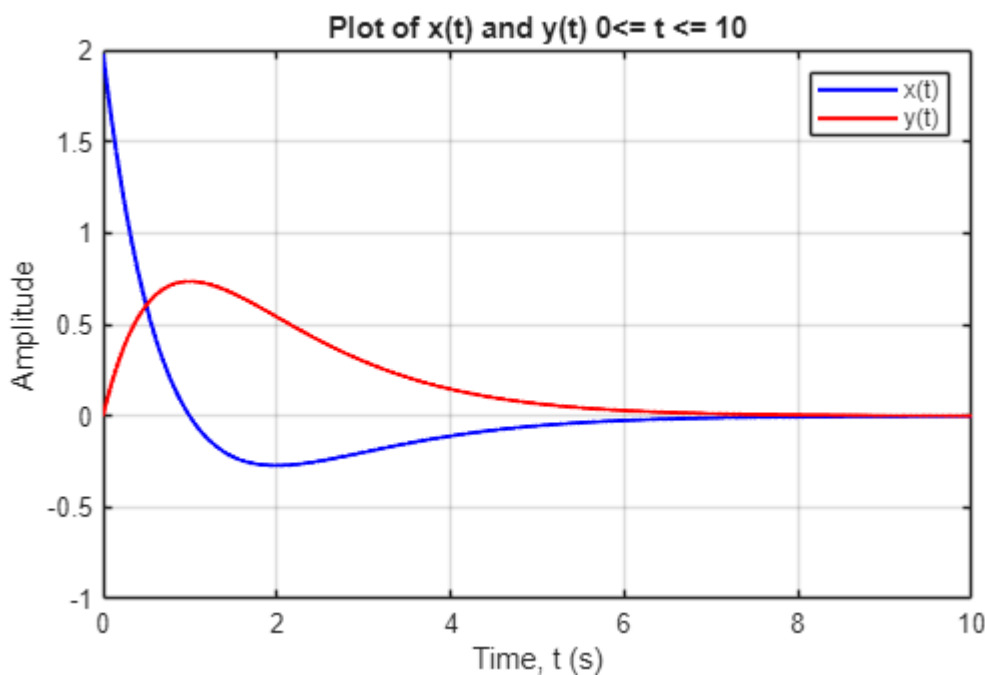


task l

```
dt = 0.01;  
t = 0:dt:10;  
y = 2.*t.*exp(-t);  
x = diff(y)/dt;
```

```
x = 1×1000  
1.9801    1.9407    1.9019    1.8636    1.8260    1.7889    1.7523    1.7163 ...
```

```
figure;  
plot(t(1:end-1), x, 'b', 'LineWidth', 1.5);  
hold on;  
plot(t, y, 'r', 'LineWidth', 1.5);  
grid on  
legend('x(t)', 'y(t)')  
xlabel('Time, t (s)');  
ylabel('Amplitude');  
title('Plot of x(t) and y(t) 0 ≤ t ≤ 10');  
axis([0 10 -1 2])
```



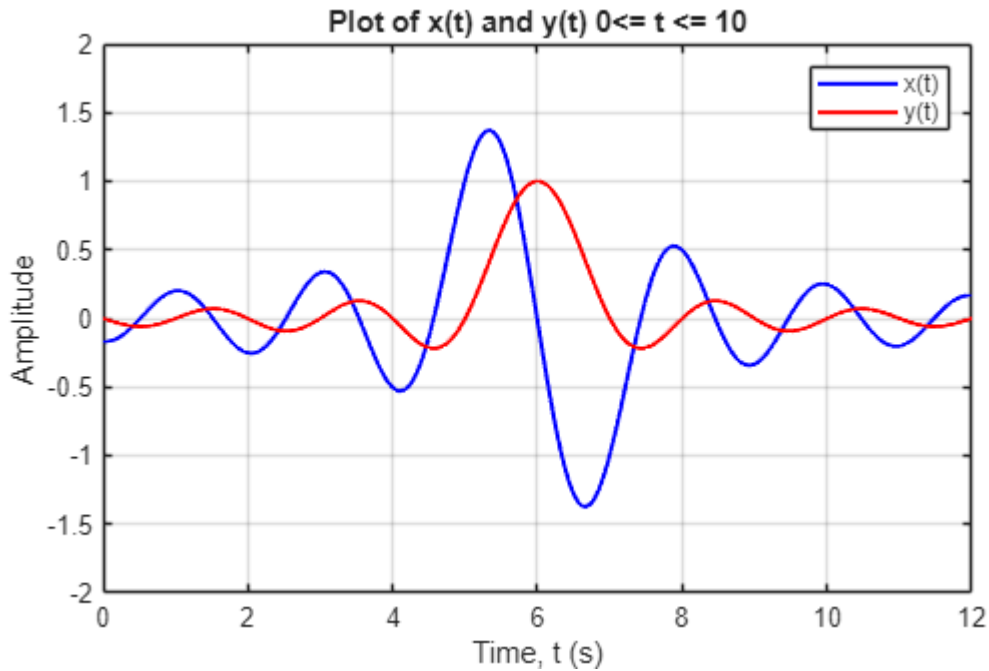
task m

```
dt = 0.01;  
t = 0:dt:12;  
y = sinc(t-6);  
x = diff(y) / dt;  
figure;  
plot(t(1:end-1), x, 'b', 'LineWidth', 1.5);
```

```

hold on;
plot(t, y, 'r', 'LineWidth', 1.5);
grid on
legend ('x(t)', 'y(t)')
xlabel('Time, t (s)');
ylabel('Amplitude');
title('Plot of x(t) and y(t) 0<= t <= 10');
axis([0 12 -2 2])

```

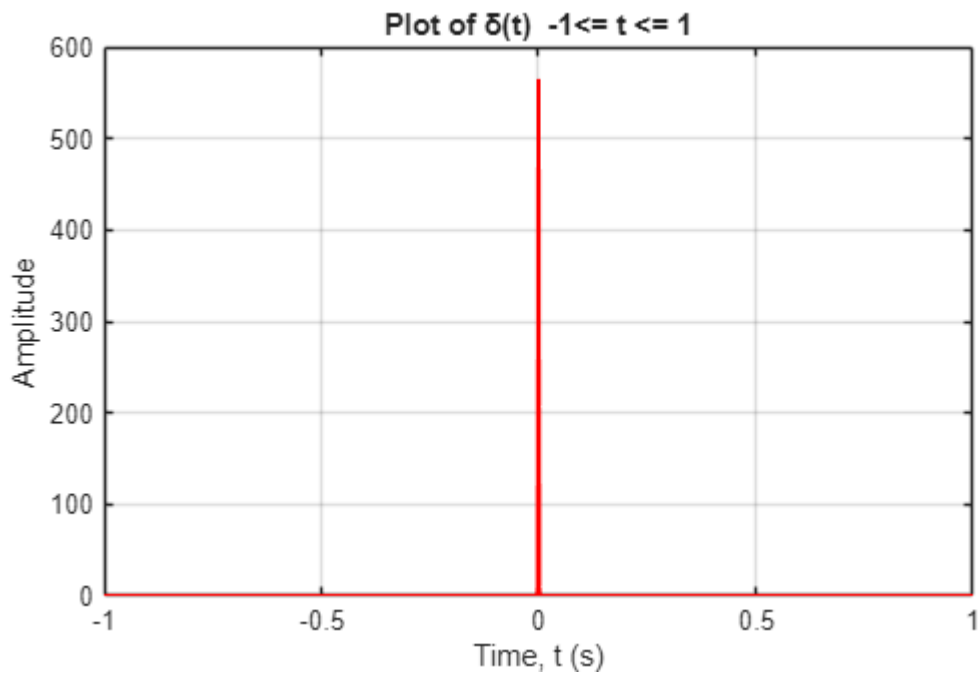


task n

```

dt = 0.001;
a = 0.001;
t = -1:dt:1;
U_impulse = 1/(a*sqrt(pi))*exp((-t.^2)/a^2);
figure;
plot(t, U_impulse, 'r', 'LineWidth', 1.5);
grid on
xlabel('Time, t (s)');
ylabel('Amplitude');
title('Plot of  $\delta(t)$  -1<= t <= 1');

```



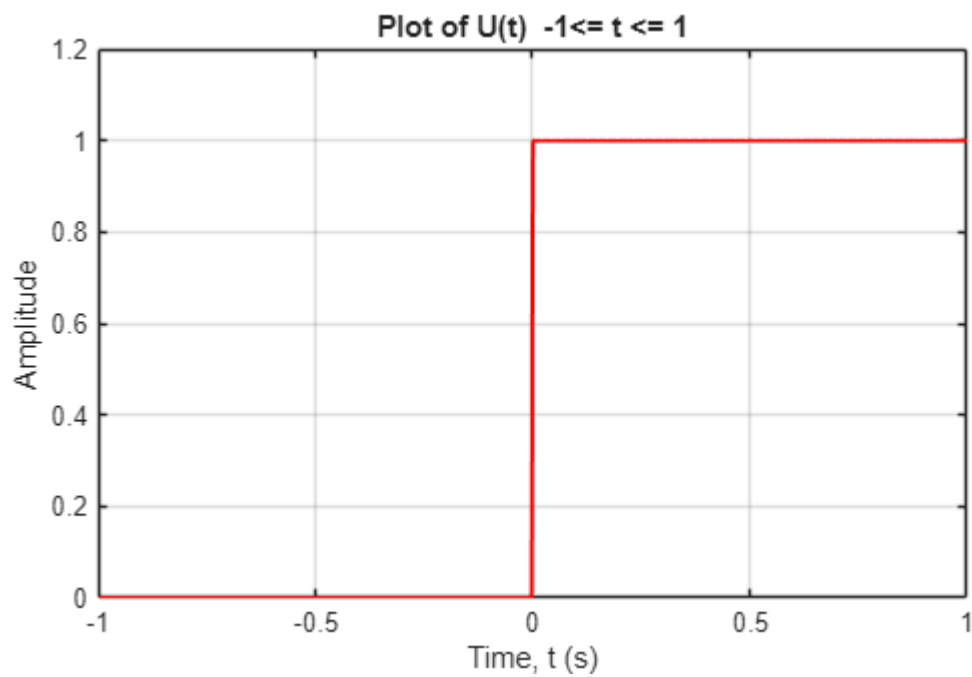
task o

```
s = sum (U_impulse);
y = dt*s
```

```
y =
1.0001
```

task p

```
c_s = cumsum(U_impulse);
u_t = dt*c_s;
plot(t, u_t, 'r', 'LineWidth', 1.5);
grid on
xlabel('Time, t (s)');
ylabel('Amplitude');
title('Plot of U(t) -1 ≤ t ≤ 1');
```



task q

```
U_imp = diff(u_t)/dt;
plot(t(1:end-1), U_imp, 'r', 'LineWidth', 1.5);
grid on
xlabel('Time, t (s)');
ylabel('Amplitude');
title('Plot of  $\delta(t)$   $-1 \leq t \leq 1$ ');
```

